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New paleomagnetic and geochronologic results from late Paleozoic rocks in the Turfan-Hami block (NW China) and implications for the geodynamic evolution of the western Altaids

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ABSTRACT

How and when the ocean-continent transition started in the western Altaids remain controversial. The paleomagnetic signals recorded by late Paleozoic rocks in the Turpan-Hami block can provide critical constraints on this issue. We conducted a new combined paleomagnetic and geochronologic study on the late Paleozoic rocks from the Turpan-Hami block. Laser ablation–inductively coupled plasma–mass spectrometry zircon U-Pb dating of volcanic beds from the Upper Carboniferous Qijiaojing and Julideneng Formations yielded ages of 313.1 ± 4.3 Ma and 309.6 ± 1.9 Ma to 308.1 ± 3.6 Ma, respectively. Meanwhile, the U-Pb age of the granite intruding the Julideneng Formation is 300.3 ± 2.4 Ma. Passing a series of fold tests, the characteristic remanent magnetization (ChRM) directions of the Qijiaojing Formation are likely primary and consistent with the Kiaman reversed superchron (ca. 319–267 Ma). However, the ChRM values of the Dananhu (Middle Devonian) and Julideneng Formations all represent reverse polarity with negative fold tests, which indicate remagnetizations related to the magmatic thermal events during the late Carboniferous. Thus, two high-quality paleomagnetic poles were obtained for the periods ca. 313–308 Ma at 44.4°N, 177.3°E ($K = 22.1$, $A_{95} = 8.0°$) and ca. 300 Ma at 47.8°N, 173.9°E ($K = 116.0$, $A_{95} = 4.8°$), respectively. Comparison with published coeval paleomagnetic poles of the blocks on both sides of the Tianshan sutures suggests that the central oceanic basin of the western Paleo-Asian Ocean (between the Siberian and Tarim blocks) had been closed since the late Carboniferous (ca. 310 Ma), apart from remnant seas. In addition, a sizeable clockwise rotation (∼58°) of the Turpan-Hami block had taken place during the early Permian, with a scissor-style shrinking of the remnant Bogda marine basin in the meantime. This study provides a new perspective for understanding the tectonic evolution of the western Altaids.

INTRODUCTION

The Altaids tectonic domain, known as the Central Asian Orogenic Belt, located between the Tarim–North China craton and the Siberian block (Fig. 1A), is the largest accretion-type orogeny on Earth, resulting from the lithospheric subduction and closure of the Paleo-Asian Ocean (Xiao et al., 2015; Şengör et al., 2018). As one of the world-renowned metallogenic provinces, the Altaids domain provides not only abundant invaluable resources but also an exciting window into Earth's mantle and crustal evolution, continental accretion, and metallogenesis. Quantifying the tectonic evolution of the Altaids is critical to understanding the amalgamation of the latest supercontinent, Pangea (Zhao et al., 2018). As the western part of the Altaids, northern Xinjiang in China recorded the critical tectonic transition from ocean closure to intracontinental orogeny between the Tarim and Siberian cratons during the late Paleozoic. It has been generally accepted that the final closure of the western Paleo-Asian Ocean occurred along the South or North Tianshan suture (Fig. 1B; Wilhem et al., 2012; Xiao et al., 2018; Zhao et al., 2018; Şengör et al., 2018). However, the final closure time of this ancient ocean is still controversial. Based on research on the geochemistry of magmatic rocks and ophiolites, the sedimentary environment, and structural deformation, etc., at least three various geologic periods have been put forward: (1) before the early Carboniferous (Dong et al., 2011; Xia and Li, 2020; Xu et al., 2014); (2) during the late Carboniferous to early Permian (Wang et al., 2006; Li et al., 2009a; Han and Zhao, 2018), or (3) the latest Permian to mid-Triassic (Xiao et al., 2015; Wang et al., 2022).

By providing quantitative constraints on the paleolatitudinal drift and vertical-axis rotation of the blocks, paleomagnetism is an effective method to address the above puzzle. The Paleozoic apparent polar wander (APW) paths for the Tarim block, located on the southern margin of the Paleo-Asian Ocean, have been reported (Zhu et al., 1998b; Zhao et al., 2014; Huang et al., 2019). In contrast, northern Xinjiang, assembled by collision/accretion of various geologic subunits (Fig. 1B), was not yet a whole block until the latest Paleozoic (Yi et al., 2015; Choulet et al., 2011; Zhu et al., 2019). These characteristics undoubtedly increase the difficulties of research in this area. Based on a Permian–Carboniferous paleomagnetic study in the Junggar region, Li et al. (1991) concluded that the Junggar block was separated from Tarim by the ancestral Tianshan oceans during the Carboniferous to Permian, and the two blocks might have converged by the late Permian (Zhao et al., 1990). New paleomagnetic results from the Yili block, the Aqishan-Yamansu belt, and western Junggar indicate that the amalgamation among the Junggar, Yili, Central Tianshan, and Tarim blocks was terminated before the latest Carbon-

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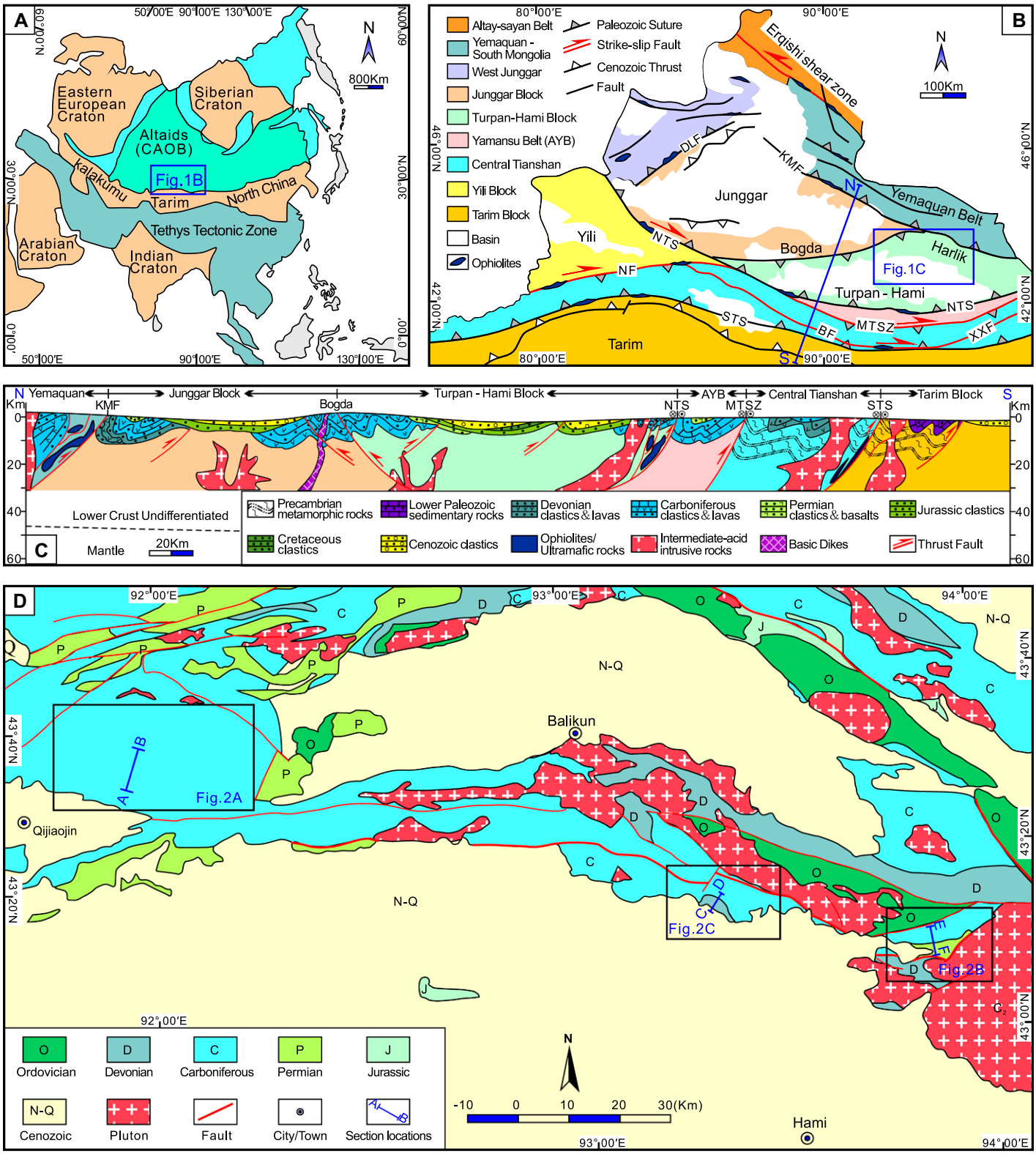


Figure 1. (A) Simplified tectonic sketch map of Asia showing the location of Chinese Tianshan (after Zhao et al., 2018). (B) Simplified tectonic division of the Chinese Tianshan, western Altaids (after Choulet et al., 2013). (C) Interpreted geophysical section from Yemaquan belt to northern Tarim (modified from Xiao et al., 2014). (D) Regional geologic map of northern Turpan-Hami block (modified from XGSC, 2007). The black boxes show the locations of sampled areas shown in Figure 2. AYB—Aqishan-Yamansu belt; BF—Baluntai fault; CAOB—Central Asian Orogenic Belt; DLF—Dalabute fault; KMF—Kelameili fault; MTSZ—Main Tianshan shear zone; NF—Nalati fault; NTS—North Tianshan suture; STS—South Tianshan suture; XXF—Xingxingxia fault.

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iferous (Wang et al., 2007; Xu et al., 2021) or during the interval between the late Carboniferous and late Permian (Yi et al., 2015). Based on the above limited paleomagnetic data, some of which were not well constrained in magnetization age or lack assessment of inclination shallowing in clastic sedimentary rocks (Meert et al., 2020), the closure time of the Paleo-Asian Ocean remains an open question. To date, we still need more robust paleomagnetic data from the late Paleozoic section of northern Xinjiang to constrain the closure timing of the western Paleo-Asian Ocean.

The western Altaids underwent remarkable intracontinental reworking regulated by late Paleozoic to Mesozoic large-scale strike-slip faulting (Laurent-Charvet et al., 2003; Wang et al., 2007; Choulet et al., 2012). Subsequently, the previous structures were further reactivated in response to the accretion and collision of India-Asia (Molnar and Tapponnier, 1975; Avouac et al., 1993), resulting in the structural architecture of the present Tianshan Range (Fig. 1C). Many Mesozoic–Cenozoic paleomagnetic data sets have quantified continental shortening and regional block rotation in northern Xinjiang under the far-field effects of the collision of the Indian and Tibetan blocks (Chen et al., 1991; Cogné et al., 1995; Huang et al., 2004a, 2004b; Choulet et al., 2013). Based on Permian–Carboniferous paleomagnetic poles, it has been proposed that the Junggar-Yili-Turpan-Hami region has gone through large counterclockwise rotations compared to the Tarim and Siberian blocks since the late Carboniferous (Wang et al., 2007; Choulet et al., 2011; Zhu et al., 2018a; Xu et al., 2021). However, the tectonic mechanism and development remain poorly understood, and further paleomagnetic studies are needed to better constrain the more precise movements of various blocks in northern Xinjiang during late Paleozoic. To date, there is no reliable late Carboniferous pole for the Turpan-Hami block, which represents a pivotal geologic unit within northern Xinjiang to research the tectonic evolution of the western Altaids (Fig. 1B) and possibly support the above supposition.

Currently, only one early Permian paleomagnetic pole of the Turpan-Hami block has been reported by Zhu et al. (2019), and authentic late Paleozoic paleomagnetic results are still extremely scarce in this region. Hence, we present new paleomagnetic and geochronologic results from late Paleozoic rocks to provide some reliable constraints on the relative paleogeographic positions and tectonic kinematic processes of the Turpan-Hami block and the closure timing of the western Paleo-Asian Ocean.

GEOLOGIC BACKGROUND AND SAMPLING

The Turpan-Hami block is triangular in shape (Fig. 1B), separated from the Junggar block by the Bogda orogenic belt to the northwest, from the Central Tianshan block by the North Tianshan suture to the south, and from the Yemaquan belt by the Kelameili fault to the northeast (Gu et al., 2001; Xiao et al., 2008). As the Turpan-Hami block is mainly covered by Mesozoic and Cenozoic sedimentary layers (Fig. 1D), our sampling was conducted along its northern margin, where Upper Paleozoic strata outcrops are abundant. The oldest strata exposed in this region are Ordovician, regarded as a folded basement mainly consisting of metamorphic lavas, clastic rocks, and marbles. Probably related to a late Caledonian orogenic event, Silurian strata are absent (Xu et al., 2014). From bottom to top, the late Paleozoic strata in the northern margin of the Turpan-Hami block are composed of the Middle Devonian Dananhu Formation (D2*d*), Lower Carboniferous Jiangbasitao Formation (C1*j*), Upper Carboniferous Qijiaojing Formation (C2*q*) and Julideneng Formation (C2*j*), and Lower Permian Kulai Formation (P1*k*). In the eastern part of the northern margin of the Turpan-Hami block (i.e., the Harlik Range), granites with late Carboniferous ages (308–295 Ma) are widely distributed (Song et al., 2018; Zhu et al., 2018b; Ni et al., 2021), emplaced in pre-Permian strata, and commonly overlain by the Kulai Formation (RGSGBX, 1970; IGSCU, 2013). This granitic magmatism was most likely associated with postorogenic extension (Song et al., 2018; Zhu et al., 2018b).

Altogether, 50 sites were sampled from three areas (Fig. 1D; Qijiaojing, Liudaogou, and Toudaogou). Generally, 10 cores were taken from each site by a portable gasoline-powered drill rig and oriented with a magnetic compass, while the local magnetic declinations were corrected by sun compasses (when possible) or the International Geomagnetic Reference Field (IGRF) value.

Qijiaojing Area

The paleomagnetic sampling section of the Qijiaojing Formation was located to the northeast of the town of Qijiaojing (Fig. 1D). This formation lies upon the Jiangbasitao Formation with an angular unconformity and is conformably overlain by the Julideneng Formation in the study area (IGSCU, 2013, p. 66). As shown in Figure 2A, the Qijiaojing Formation constitutes a series of fold structures with nearly E-W strikes and subhorizontal axes (plunging 4° toward 261°). Its lower part consists of dark-green basalts, light gray-green rhyolites, gray-green tuff, gray-yellow sandstones, and dark-gray silty mudstones, while the upper part is mainly composed of fine-grained sandstones and dark-gray siltstones with various thicknesses from 30 cm to 1 m (Fig. S1A[1]). The Qijiaojing Formation was considered to have a late early Carboniferous age (XGSC, 2007). However, based on the new fossils discoveries, such as *Dictyociostus* sp., *Linoproductus* sp., *Camarotoechia* sp., *Calamites* sp., and *Neoggerathiopsis* sp., Chu et al. (2016) reported that the age of the Qijiaojing Formation should be late Carboniferous. To further constrain the age of the Qijiaojing Formation, two fresh volcanic samples were collected for geochronologic dating: rhyolites from the lower section of the Qijiaojing Formation (16TEDQ-Tw1; Fig. S1D) and tuff (16TL-Tw1) from the lower section of the Julideneng Formation. In addition, we sampled six siltstone (Fig. S1B) sites, five sandstone sites, and ten basalt (Fig. S1C) sites from the Qijiaojing Formations (Fig. 2A).

Toudaogou Area

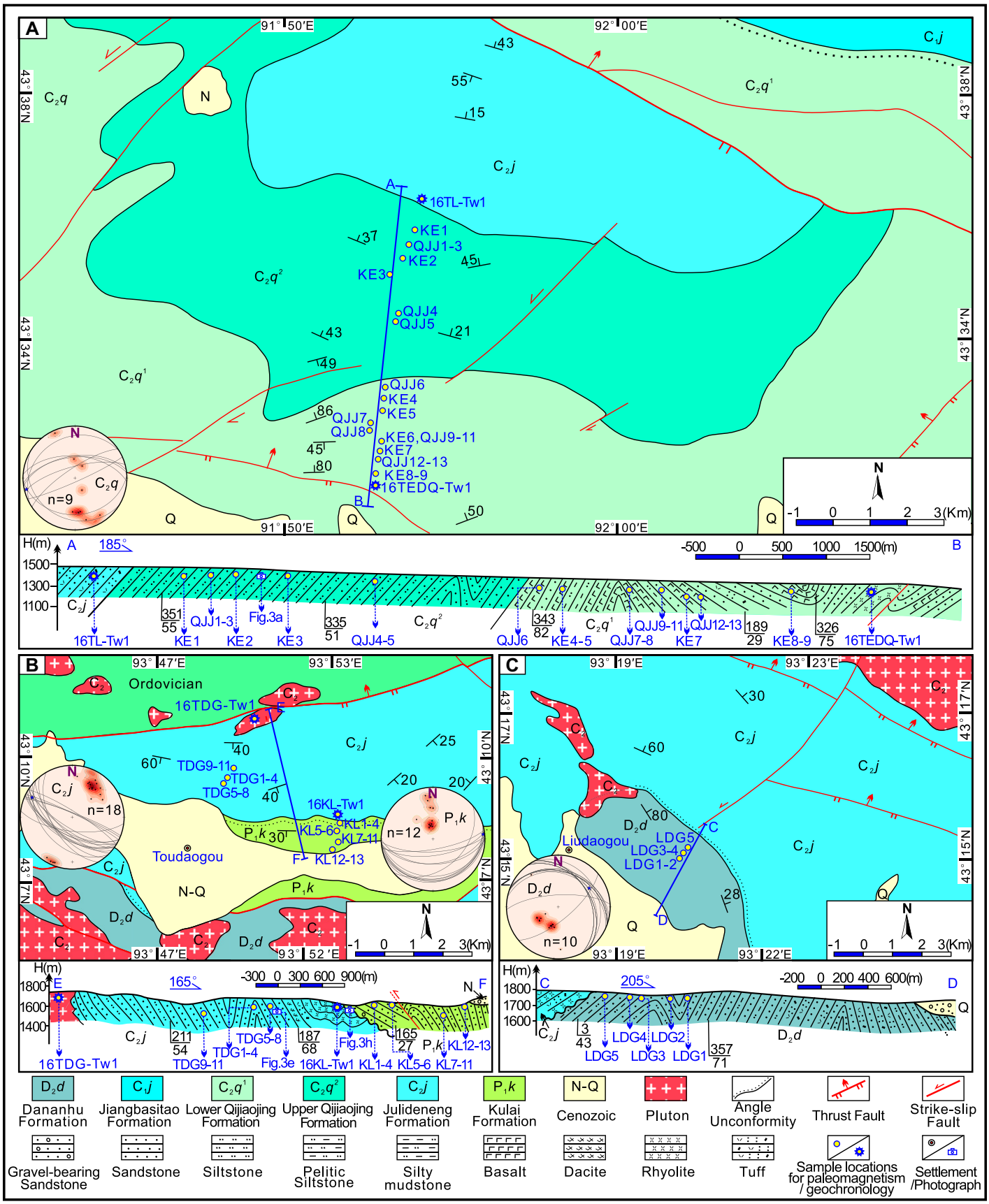
The Julideneng and Kulai Formations were sampled in the Toudaogou area located to the northeast of Hami City (Fig. 1D). The Julideneng Formation consists of gray sandstones, dark-gray siltstones, gray-green tuff, light gray-green dacites, and rhyolites, forming NE-SW–striking folds with subhorizontal axes (Fig. 2B). The Kulai Formation is mainly composed of gray-yellow sandstones, and it lies on the Julideneng Formation with an angular unconformity (Fig. S1H). The ages of the Julideneng Formation and Kulai Formation are regarded as late Carboniferous and early Permian, respectively, according to abundant fossils found in the strata (RGSGBX, 1970, p. 10–14). Altogether 11 and 13 sandstone sites were sampled from the Julideneng (Figs. S1E–S1F) and Kulai Formations (Fig. S1I), respectively. To further refine the age of the Julideneng Formation, a fresh rhyolite sample (16KL-Tw1; Fig. S1G) from the top of the formation and a granite sample (16TDG-Tw1) intruding into it were collected for geochronologic dating.

Liudaogou Area

The Dananhu Formation, unconformably overlain by the Julideneng Formation, was sam-

[1]Supplemental Material. Methods (Texts S1–S3), Tables S1–S3, and Figures S1–S4. Please visit https://doi.org/10.1130/GSAB.S.23800632 to access the supplemental material, and contact editing@geosociety.org with any questions.

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Figure 2. Geologic maps and sections of (A) Qijiaojing, (B) Toudaogou, and (C) Liudaogou areas (modified from IGSCU, 2013; RGSGBX, 1970). Lower-hemisphere stereographic projections of the sampled site bedding attitudes determine the fold axes plunging 4° toward 261° for the Qijiaojing Formation (A), plunging 9° toward 295° for the Julideneng Formation and plunging 0.3° toward 84° for the Kulai Formation (B), and plunging 31° toward 74° for the Dananhu Formation (C). Measured geologic sections show the details about sampling locations, outcrop photographs (shown in Fig. S1 [see text footnote 1]) positions, and bedding attitudes.

pled in the Liudaogou section located ~50 km northwest of Hami City (Fig. 1D). The sampling section constituted a series of folds with inclined axes (plunging 31° toward 74°) and lithology consisting of gray-yellow sandstones, gray-green tuff, dark-gray pelitic siltstones, and silty mudstones (Fig. 2C; Fig. S1J). Based on the marine fossil evidence, the age of this formation is generally suggested as Middle Devonian (RGSGBX, 1970, p. 5–8). Four pelitic siltstone sites and one sandstone site were sampled in the Liudaogou area.

METHODS

The scanning electron microscope (SEM) observations, energy-dispersive spectroscopy (EDS) analyses, and zircon U-Pb isotope analyses were carried out in the Key Laboratory for the Study of Focused Magmatism and Giant Ore Deposits, China Geological Survey (Geoscience Innovation Center of Northwest China), Xi'an, China. A JSM-6510A SEM was used to conduct SEM observations and EDS analyses on representative samples. The zircon U-Th-Pb analyses were obtained on an Agilent (7700X) inductively coupled plasma–mass spectrometer (ICP-MS) with a laser-ablation microprobe (GeoLas Pro). Detailed analytical methods for U-Pb zircon dating, SEM observations, and EDS analyses are available in Supplemental Text S1.

All rock magnetic experiments and systematic demagnetization were conducted in the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Natural Resources (MNR), Chinese Academy of Geological Sciences, Beijing, China. A Lake Shore vibrating sample magnetometer (VSM, 8600 series) was used to obtain the isothermal remanent magnetization (IRM) acquisition curves and hysteresis loops for representative samples. A KLY-3 bridge (AGICO) was used to measure the anisotropy of magnetic susceptibility (AMS) and thermomagnetic curves of susceptibility of specimens. Stepwise thermal demagnetization was conducted using an ASC-TD-48 thermal demagnetizer, and remanent magnetization was measured by a 2G superconducting magnetometer. The alternating-field (AF) demagnetization was performed using a U-Channel rock superconducting magnetometer system. Detailed systematic demagnetization methods are supplied in Supplemental Text S2. Paleomagnetic data were analyzed by paleomagnetism.org (Koymans et al., 2020) and the software package of Enkin (2003) and Cogné (2003), and the mean paleomagnetic directions were calculated using Fisher statistics (Fisher, 1953).

RESULTS

Zircon U-Pb Ages

The U-Pb isotope analyses for zircons from the samples (16TEDQ-Tw1, 16TL-Tw1, 16KL-Tw1, and 16TDG-Tw1) are shown in Table S1 and Figure 3. All zircons from the samples showed high Th/U ratios (between 0.3 and 1.8) with clear concentric oscillatory zoning in their cathodoluminescence (CL) images (Figs. 3C, 3F, 3I, and 3L), which are characteristics of magmatic zircons (Wu and Zheng, 2004).

16TEDQ-Tw1

Thirty-six zircons were analyzed from 16TEDQ-Tw1, and 26 analyses were ≤10% discordant (Fig. 3A). The youngest age group (22 concordant ages) yielded a weighted mean $^{206}Pb/^{238}U$ date of 313.1 ± 4.3 Ma (Figs. 3A and 3B), which is considered to be the crystallization age of the rhyolite. The other four zircons with older ages (396, 352, 350, and 332 Ma) mostly showed inherited cores with reaction rims (Fig. 3C, spots 13, 16, 22), indicating that they are xenocrysts from the magmatic source or volcanic conduit.

16TL-Tw1

Thirty-six zircons were also analyzed from 16TL-Tw1, and 34 analyses were ≤10% discordant (Fig. 3D). The age distribution showed apparent peaks at 440 Ma, 407 Ma, 330 Ma, and 308 Ma (Fig. 3D). The youngest age group (11 concordant ages) yielded a weighted mean age of 308.1 ± 3.6 Ma (Fig. 3E). Given that the corresponding zircons showed euhedral shapes without corrosion textures (Fig. 3F, spots 5, 8, 20, 34), this age is considered to be the formation time of the tuff. The other zircons with older ages (≥318 Ma) mostly were subrounded in shape with reaction rims (Fig. 3F, spots 11 and 19), so they are interpreted as xenocrysts crystallized during other older magmatic events. Thus, the deposition age of the sampled section in the Qijiaojing area is between 313.1 Ma and 308.1 Ma, which coincides with the biostratigraphic evidence (Chu et al., 2016).

16KL-Tw1

Zircons from sample 16KL-Tw1 mainly showed euhedral shapes (Fig. 3I). Thirty-six spots were analyzed, and 31 analyses are ≤10% discordant (Fig. 3G). All the 31 results yielded similar $^{206}Pb/^{238}U$ ages of 313–306 Ma, with a weighted mean age of 309.6 ± 1.9 Ma (mean square of weighted deviates [MSWD] = 0.118, $n = 31$; Fig. 3H), which is considered to be the eruption age of the rhyolite. Thus, the age of the Julideneng Formation should be Moscovian (late Carboniferous), which is in accord with the marine fossils in the formation (RGSGBX, 1970).

16TDG-Tw1

Eighteen spots were analyzed for sample 16TDG-Tw1, and 15 analyses were ≤10% discordant (Fig. 3J). One inherited zircon with an older age (326 Ma) was identified (Fig. 3L, spot 3). The other 14 results (with similar apparent $^{206}Pb/^{238}U$ ages of 305–296 Ma) yielded a weighted mean age of 300.3 ± 2.4 Ma (MSWD = 0.24, $n = 14$; Fig. 3K), which is interpreted as the crystallization age of the granite.

Rock Magnetic Measurements

The thermomagnetic (k-T) curves for the representative samples of the Qijiaojing Formation showed a sharp drop of magnetic susceptibility at ~585 °C, indicating the presence of magnetite (Figs. 4A and 4D). In addition, basalt sample 16TEDQ-B20 provided evidence for a Hopkinson peak near 580 °C (Fig. 4A), indicative of single-domain magnetite (Dunlop and Ozdemir, 1997). As for the samples from the Julideneng Formation (Fig. 4G), the thermal demagnetizations of three-component IRM revealed that the levels of the soft component, unblocked until ~580 °C, were significantly higher than the hard and medium components, indicating that low-coercivity magnetite grains dominated the samples from the Julideneng Formation. IRM acquisition curves of the representative samples from the Qijiaojing and Julideneng Formations showed a rapid increase below 200 mT and complete saturation at no more than ~300 mT (Figs. 4B, 4E, and 4H), suggesting the dominance of soft magnetic minerals in these samples. The results of hysteresis loop measurements for these representative samples are shown in Figures 4C, 4F, and 4I, which generally show narrow shapes

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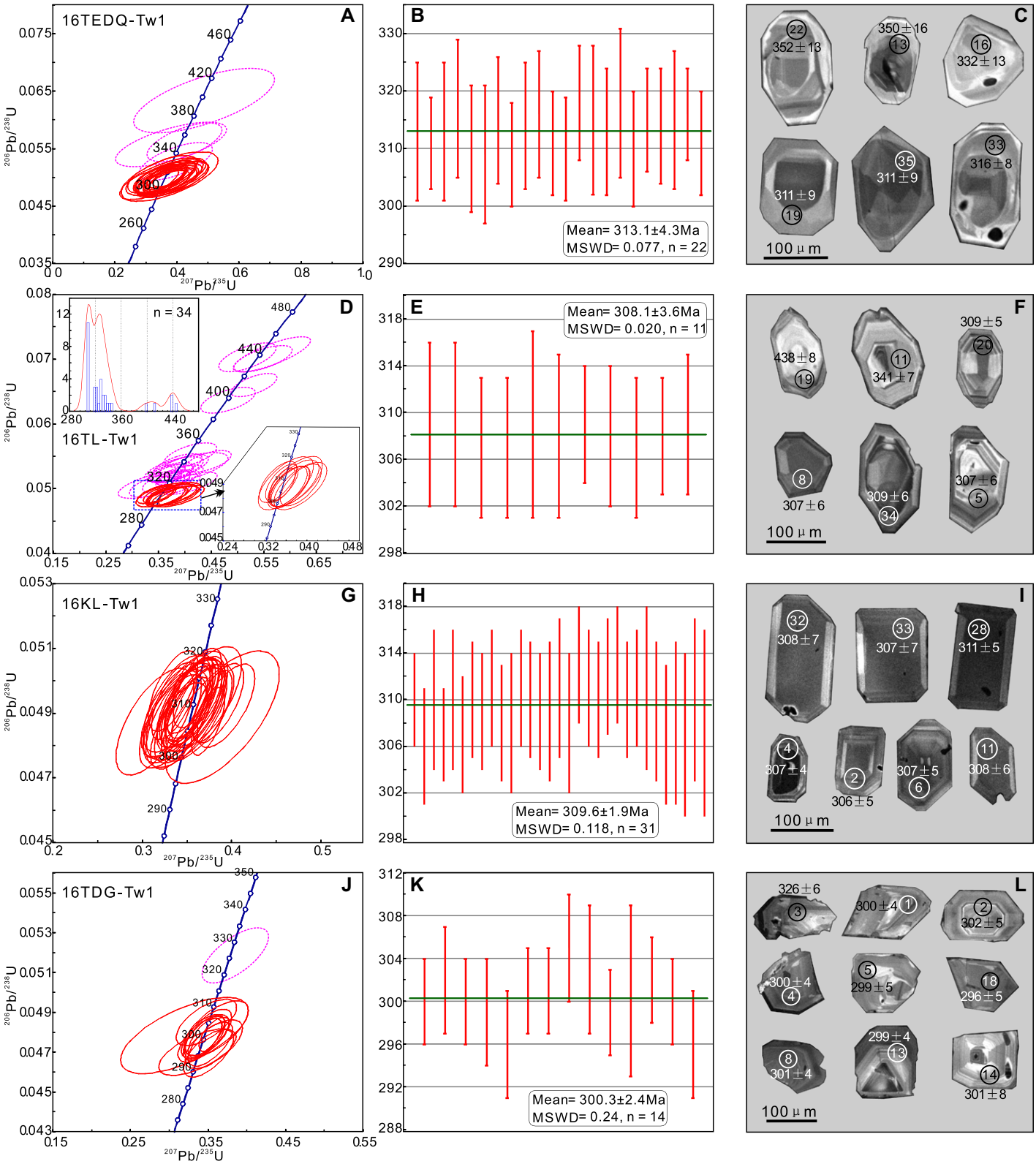


Figure 3. U-Pb concordia diagrams, weighted mean ages, and cathodoluminescence (CL) images for zircon analyses of samples: (A–C) 16KEDQ-Tw1, (D–F) 16TL-Tw1, (G–I) 16KL-Tw1, and (J–L) 16TDG-Tw1. Purple circles in the concordia diagrams show the U-Pb concordant ages of the inherited zircons. Small circles in CL images show the locations of zircon U-Pb dating with their corresponding spot number and zircon U-Pb age. MSWD—mean square of weighted deviates.

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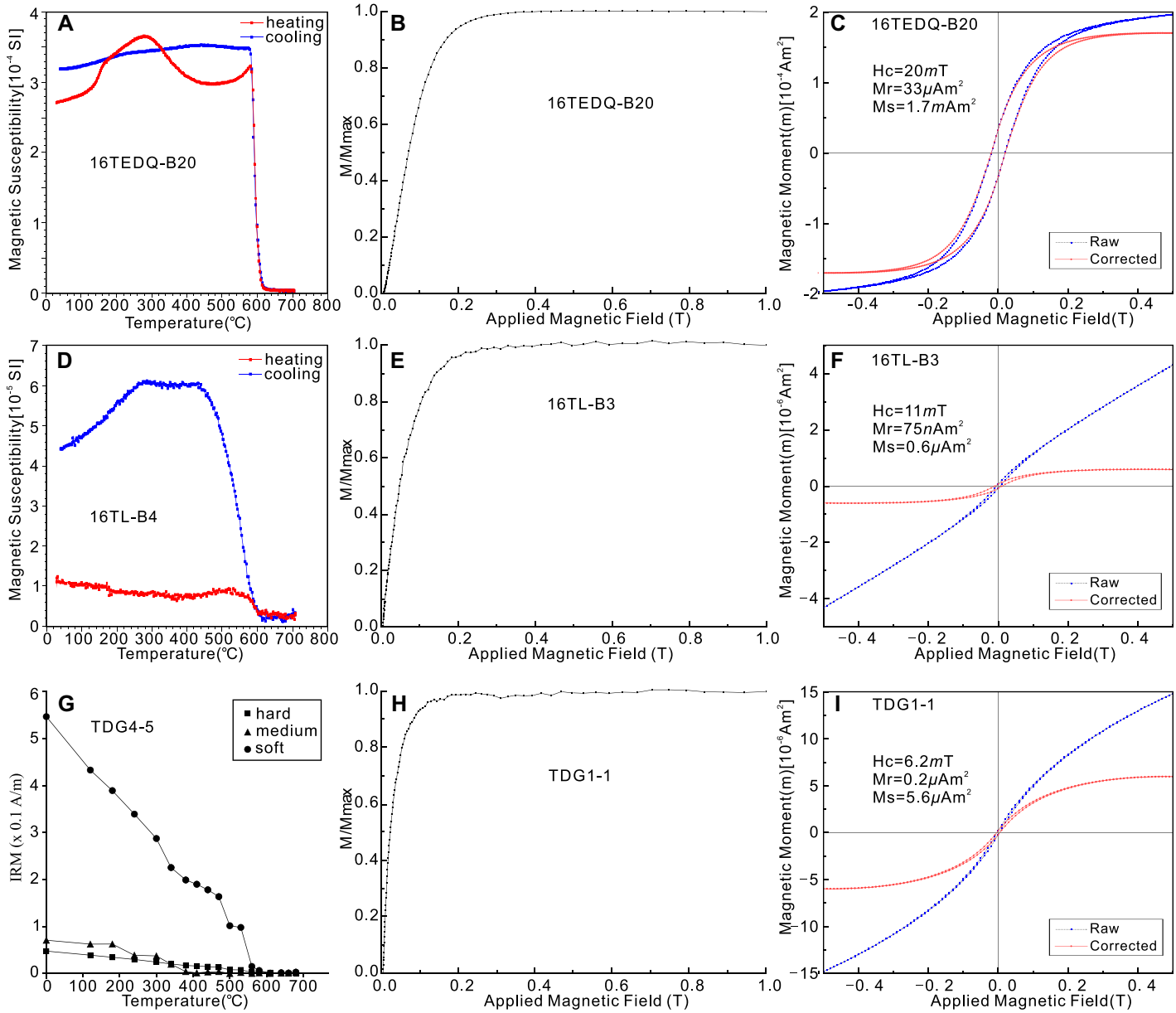


Figure 4. Magnetic mineral analyses: (A, D) thermomagnetic curves, (B, E, H) isothermal magnetization (IRM) acquisition curves, (C, F, I) magnetic hysteresis loops, and (G) thermal demagnetization of a three-component IRM of representative samples. Parts A–C and D–F show typical examples of basalt and siltstones from the Qijiaojing Formation, respectively; parts G–I show typical examples of sandstones from the Julideneng Formation. Hc—coercive force; Mr—saturation remanent magnetization; Ms—saturation magnetization.

and tend to close before 300 mT with low-coercivity values around 6–20 mT, also indicating that soft magnetic minerals (e.g., magnetite or Ti-magnetite) were predominant in the samples.

The AMS of representative specimens was measured to investigate the degree of deformation in these studied sections. The corrected P′ (anisotropy degree of AMS) values of the specimens mainly varied from 1 to 1.06, while the value of one specimen from the Kulai Formations exceeded 1.06 (Fig. S2). The anisotropy shape factors (T) of the sedimentary specimens (siltstones, sandstones) from the Qijiaojing, Julideneng, and Kulai Formations mainly ranged from 0 to 1 (Figs. S2C, S2I, and S2L), indicating the development of foliation with a dominance of oblateness. After tilt correction, K_1s and K_2s (maximum and medial susceptibility axes) were distributed randomly on the horizontal surface, and K_3s (the minimum susceptibility axes) clustered to be perpendicular to the bedding plane (Figs. S2B, S2H, and S2K), which imply that the oblate magnetic fabric was related to the primary sedimentary process. Otherwise, the principal anisotropy directions were irregular for the basalts from the Qijiaojing area (Figs. S2A and S2B). Considering such lower degrees of anisotropy, we concluded that neither the clastic rocks nor the lavas above have undergone prominent strain caused by compaction or tectonic stresses.

Instead, the T values of the sedimentary specimens from the Dananhu Formation mainly range from −1 to 0 (Fig. S2F), indicating the development of lineation. Furthermore, the K_1s are concentrated with the K_2s and K_3s distributed along a great circle

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in the in situ coordinates. Their mean direction is roughly parallel to the fold axis of the sampled section (Fig. S2D), indicating that these sedimentary rocks have experienced strain triggered by tectonic stresses during the folding stage. However, the low P′ values still indicate weakly deformed samples, consistent with field observations (Fig. S1J). The primary remanence direction should not be deflected strongly (Kligfield et al., 1983; Gao et al., 2014).

SEM Observations and EDS Analyses

SEM observations were conducted on representative samples of the Qijiaojing and Julideneng Formations. The magnetic minerals from representative samples mainly showed granular shapes and sizes ranging from a few microns to more than 100 μm (Figs. 5A–5F). The EDS analyses indicated that the magnetic minerals mainly consisted of iron oxide in these sedimentary samples (Figs. 5G–5H and 5K–5L). Based on the results of rock-magnetic analyses, SEM observations, and EDS analyses, we infer that magnetite predominates in the siliciclastic samples of the Qijiaojing and Julideneng Formations. In addition, the magnetic minerals of the basalts from the Qijiaojing Formation showed structures with white stripes alternating with pale ones (Figs. 5C–5D), and the EDS analyses revealed that the white and pale bars mainly consist of iron oxide and iron-titanium oxide, respectively (Figs. 5I–5J). Thus, it is suggested

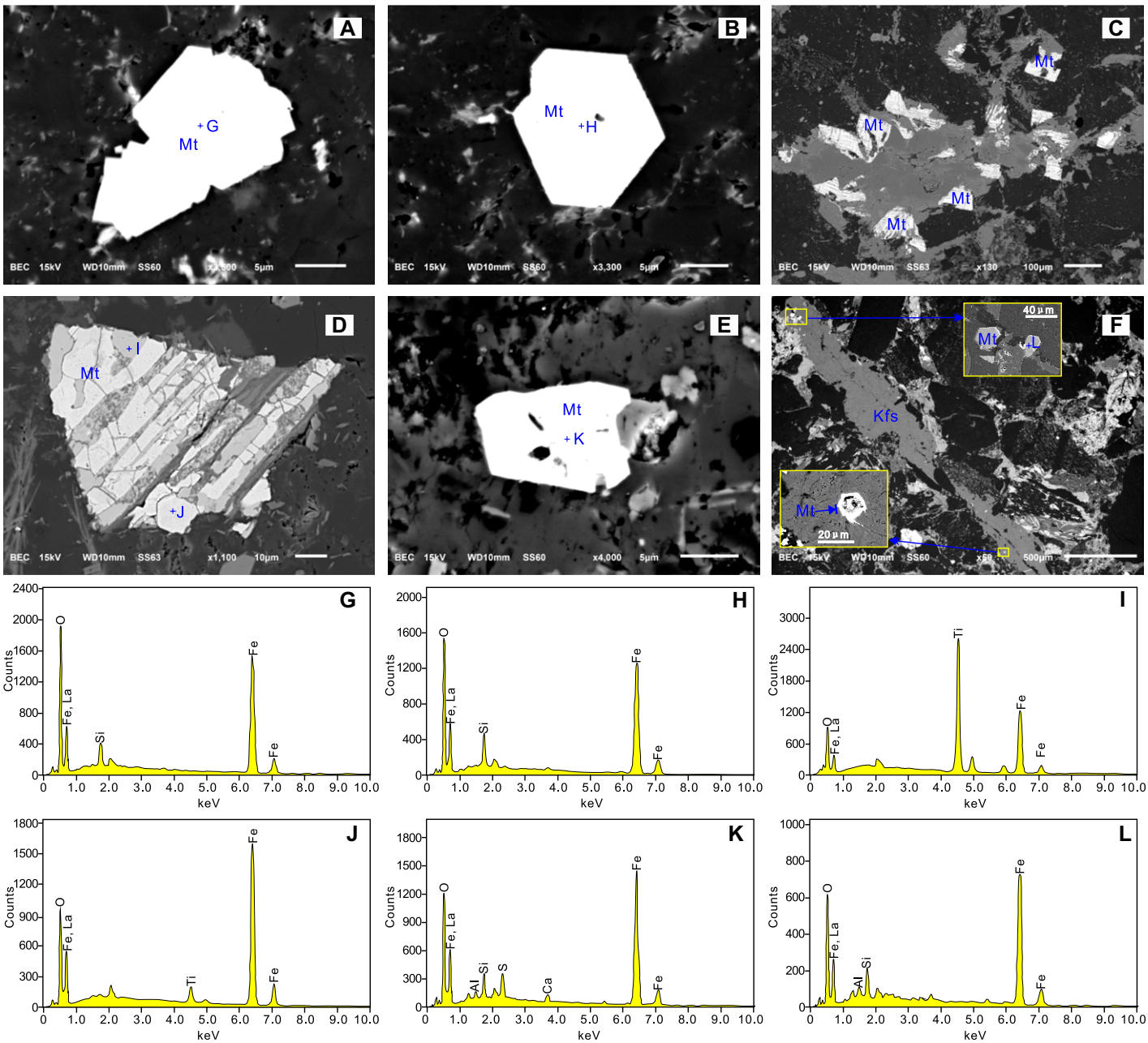


Figure 5. Representative scanning electron microscope (SEM) backscattered-electron images of samples from the (A–D) Qijiaojing and (E–F) Julideneng Formations. Small blue crosses denote the energy-dispersive spectroscopy (EDS) analysis points with the corresponding results (G–L). Mt—magnetite; Kfs—K-feldspar.

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that magnetite and ilmenite dominate in the basalt samples, and the magnetic minerals were produced by exsolution during the cooling of the lava. The magnetic minerals of samples from the Qijiaojing Formation were not affected by later veins, showing very distinct boundaries with the groundmass (Figs. 5A–5D), which suggests that these magnetic minerals were not influenced by chemical alteration caused by later fluids or hydrothermal activity. The above-listed evidence indicates that the magnetic minerals of the samples from the Qijiaojing Formation should be primary. Although some magnetites in the samples from the Julideneng Formation present a primary detrital character (Fig. 5E), there are also some secondary magnetites in several later quartz and K-feldspar veins in these samples (Fig. 5F).

Paleomagnetic Directions

Acquisition of Characteristic Remanent Magnetization (ChRM)

Multiple demagnetization methods were used to confirm the stepwise demagnetization effectiveness. In contrast, most specimens (246) were stepwise treated by thermal demagnetization, which gave better results than other methods. For a small proportion of specimens (29), we used a mixture of thermal and AF demagnetization or single AF demagnetization. The paleomagnetism directions obtained by the different demagnetization methods were roughly consistent. Figure 6 shows the orthogonal vector diagrams of representative specimens. Two components were revealed for most of the specimens: a low-stability component (LSC) isolated before 240–470 °C or 7.5–20 mT, and a characteristic remanent magnetization (ChRM) isolated between 240–470 °C and 580 °C or 5–50 mT

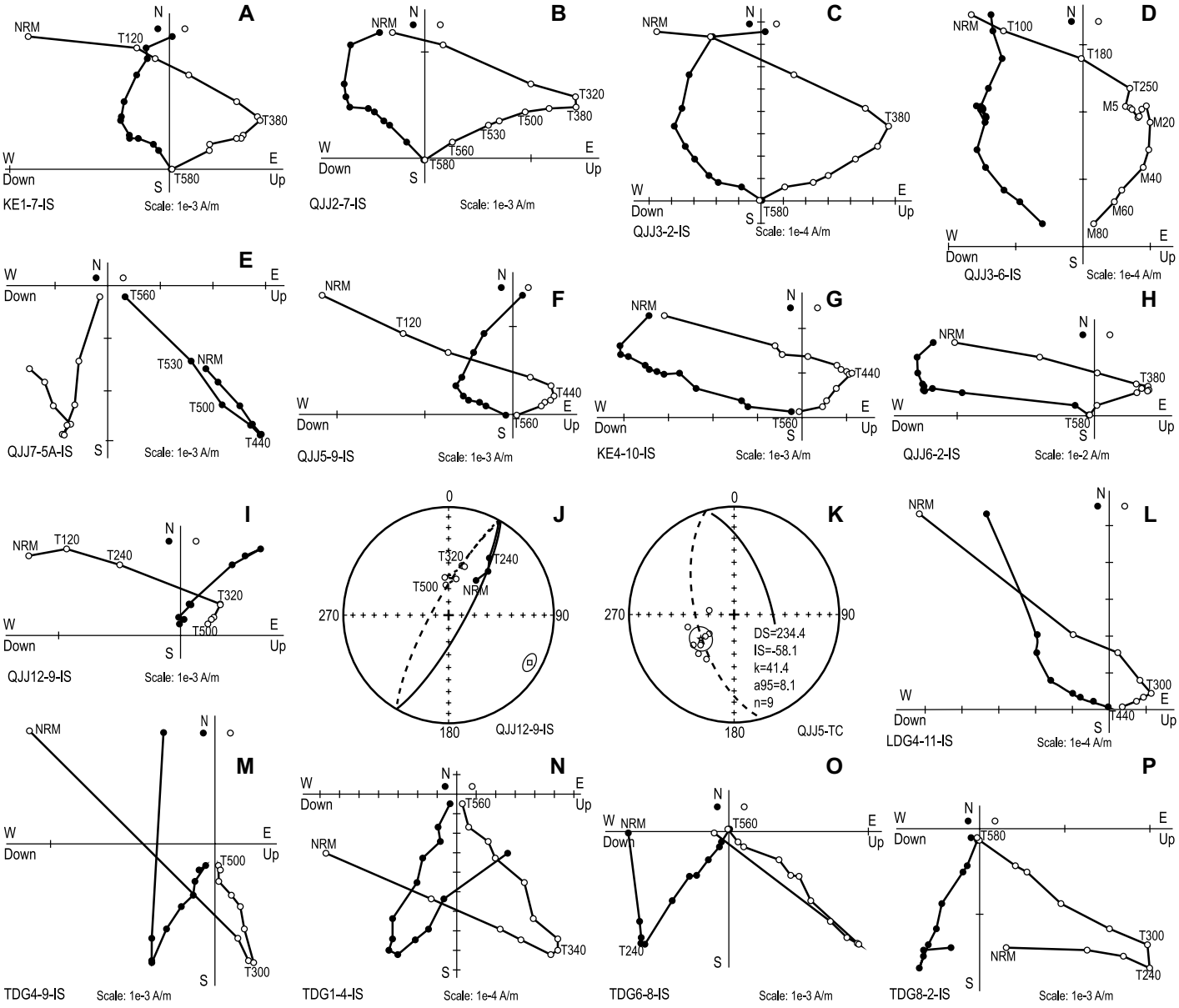


Figure 6. Representative orthogonal demagnetization diagrams for the specimens from (A–J) Qijiaojing Formation, (L) Dananhu Formation, and (M–P) Julideneng Formation in the in situ coordinates. (K) QJJ5 site-mean characteristic remanent magnetization (ChRM) direction (open star) in tilt-corrected coordinates with α_{95} (small circle). Solid (open) circles represent horizontal (vertical) projections. NRM—natural remnant magnetization.

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and 80–90 mT for thermal demagnetization and AF demagnetization, respectively. ChRMs usually decayed toward the origin (Figs. 6A–6H and 6L–6P), and the directions were analyzed using principal component analysis (Kirschvink, 1980). Moreover, the ChRMs of seven specimens from the Qijiaojing Formation were distributed along a great circle (Figs. 6I–6J), which could be analyzed by remagnetization circles analysis (see Fig. 6J; Halls, 1978). In this case, the corresponding paleomagnetic site-mean directions were calculated with the mixed mean of the principal components and remagnetization great circles (see Fig. 6K; McFadden and McElhinny, 1988). As the sampled section of the Dananhu Formation constituted plunging folds (Fig. 2C), tilt correction with an inclined fold axis was also conducted for the ChRM directions.

Statistical Results

The site-mean directions of the LSC are listed in Table S2 and shown in Figures 7A–7B. With the exceptions of sites KE3 and KL7, the site-mean direction of the LSC for the other sampling sites was $D_g = 6.8°$, $I_g = 64.9°$, $k_g = 65.6$, $\alpha_{95} = 2.7°$, in geographic coordinates, and $D_s = 358.8°$, $I_s = 73.5°$, $k_s = 2.5$, $\alpha_{95} = 18.1°$ in stratigraphic coordinates, with $N = 42$. Specimen-level ChRM directions were calculated to virtual geomagnetic poles (VGPs) and filtered with a fixed 45° cutoff (Fig. S3; Deenen et al., 2011). The specimen-level and site-mean directions of the ChRMs are listed in Table S3 and Tables 1–2, respectively. The specimen-mean ChRM direction of the Qijiaojing Formation in the Qijiaojing area is $D_g/I_g = 271.4°/-44.9°$, $k_g = 4.5$, $\alpha_{95} = 7.2°$ before and $D_s/I_s = 238.6°/-52.1°$, $k_s = 11.7$, $\alpha_{95} = 4.1°$ after tilt correction with $n = 111$; that of Julideneng Formation in the Toudaogou area is $D_g/I_g = 210.1°/-40.4°$, $k_g = 42.9$, $\alpha_{95} = 3.0°$ before and $D_s/I_s = 162.8°/-83.6°$, $k_s = 2.5$, $\alpha_{95} = 16.0°$ after tilt correction with $n = 54$ (Table S3). In addition, the site-mean ChRM direction of the Qijiaojing Formation is $D_g/I_g = 262.5°/-41.6°$, $k_g = 4.9$, $\alpha_{95} = 18.6°$ before and $D_s/I_s = 236.6°/-50.3°$, $k_s = 33.5$, $\alpha_{95} = 6.5°$ after tilt correction with $N = 16$ (Figs. 7D and 7E; Table 1); that of the Julideneng Formation is $D_g/I_g = 209.4°/-40.4°$, $k_g = 181.7$, $\alpha_{95} = 3.8°$ before and $D_s/I_s = 151.8°/-81.5°$, $k_s = 2.3$, $\alpha_{95} = 44.4°$ after tilt correction with $N = 9$ (Figs. 7G and 7H; Table 2). The above specimen- and site-level mean directions are indistinguishable at a 95% confidence level, and the site-mean directions were used in the following analysis.

The ChRM directions of specimens from the Kulai Formation in the Toudaogou area were scattered over a wide region with no statistical significance. Although 26 specimens of the Dananhu Formation in the Liudaogou area were stepwise treated by thermal demagnetization, stable ChRMs were only obtained from five of them. The specimen-mean direction of the ChRMs is $D_g/I_g = 252.6°/-46.6°$, with $k_g = 15.8$, $\alpha_{95} = 19.8°$, and $n = 5$ in geographic coordinates (Fig. 7J), $D_s/I_s = 234.9°/12.1°$, with $k_s = 10.5$, $\alpha_{95} = 24.8°$ after tilt correction (Fig. 7K), and $D_{st}/I_{st} = 246.7°/-12.0°$, with $k_s = 11.8$, $\alpha_{95} = 23.8°$ after tilt correction with inclined fold axis (declination = 74°; inclination = 31°).

DISCUSSION

Interpretation of the Magnetic Remanence

The LSC direction is characterized by a negative paleomagnetic fold test with a test statistic maximized at $-1.2\% \pm 2.3\%$ unfolding (Figs. 7A–7C; McElhinny, 1964; Watson and Enkin, 1993), indicating that the LSC was obtained after folding. Considering that the LSC direction in geographic coordinates is almost consistent with the direction of the present geomagnetic field (PGF) (Fig. 7A) at the mean sampling location (43.4°N, 93.0°E), it is inferred that the LSC direction represents recent overprints. In contrast, the ChRM directions are all obviously different from the PGF, Cenozoic, Mesozoic, or early Permian geomagnetic directions (Figs. 7D, 7G, and 7J; Cogné et al., 1995; Huang et al., 2004a, 2004b; Zhu et al., 2019), which might imply that these ChRMs were acquired before the early Permian. However, each sampled section should be distinguished correctly to check whether a remanence is primary or remagnetization.

Qijiaojing Area

The ChRM directions of the Qijiaojing Formation in the Qijiaojing area passed the McElhinny (1964) fold test at 95% and 99% confidence levels ($k_s/k_g = 6.81 > 1.84$ or 2.38). The McFadden (1990) fold test showed that $\xi(1)_{\text{in situ}} = 13.99$, $\xi(1)_{\text{tilt-corrected}} = 3.58$, and $\xi(2)_{\text{in situ}} = 14.86$, $\xi(2)_{\text{tilt-corrected}} = 0.114$, while the critical value was $\xi C = 4.66$ and 6.52 at 95% and 99% confidence levels, respectively, indicating a positive fold test at 95% and 99% confidence levels. Furthermore, the analysis of stepwise unfolding illustrated that the site-mean direction of the ChRM generated a maximum concentration at $\sim 93.1\% \pm 4.2\%$ unfolding (Fig. 7F; Watson and Enkin, 1993), and direction-correction (DC) slope = 0.936 ± 0.142 (Enkin, 2003), which also indicates a positive fold test. Thus, the age of magnetization is prefolding. The angular unconformity between the Upper Carboniferous and the Lower Permian strata is widely distributed throughout the studied region (IGSCU, 2013; RGSGBX, 1970; see also Fig. S1H), so the magnetization should have been obtained before the early Permian. In addition, most of the ChRM directions represent reverse polarity (normal = 2, reverse = 102) after tilt correction, which coincides with the Kiaman reversed superchron (ca. 319–267 Ma; Hounslow, 2022). Moreover, the SEM observations illustrated that either basalts or siltstones of the Qijiaojing Formation contained primary magnetic minerals without having undergone chemical alteration. All these characteristics suggest that the CHRM of the Qijiaojing Formation represents original remanent magnetization.

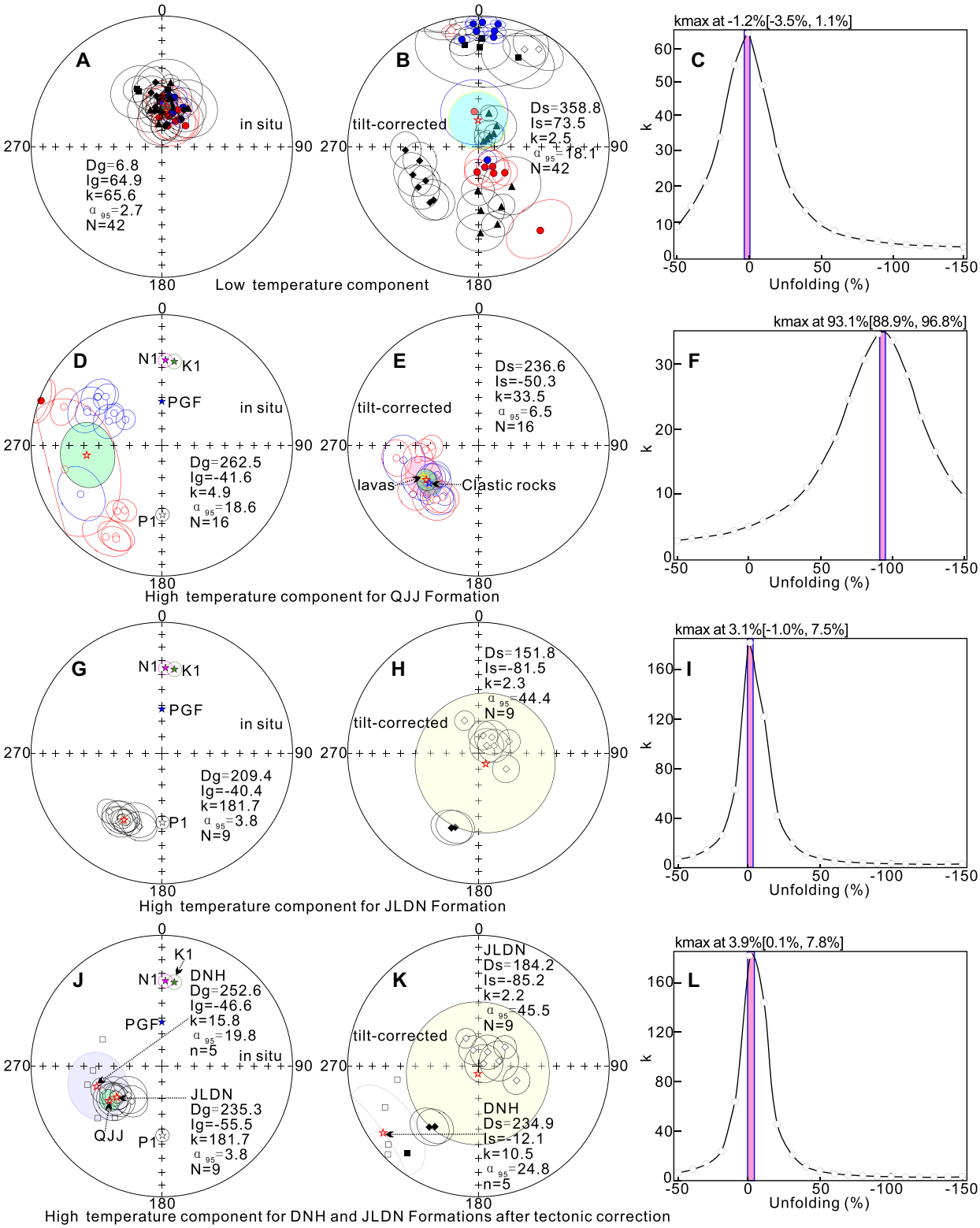
Figure 7. Equal-area projections and progressive unfolding analyses of the low-stability component (LSC) for all sites (A–C), the site-mean characteristic remanent magnetization (ChRM) directions for Qijiaojing (QJJ) (D–F) and Julideneng (JLDN) (G–L) Formations as well as the Dananhu (DNH) Formation (J–K). Open (filled) symbols denote upward (downward) inclinations. PGF—present geomagnetic field.

The extent of inclination shallowing is directly related to the degree of magnetic anisotropy (Jackson et al., 1991). Based on the low P′ values (varying from 1.017 to 1.047) within the sedimentary specimens and consistent inclination recorded in the clastic rocks and volcanic lavas ($I_{\text{sediment}} = -50.7 \pm 7.5°$, $I_{\text{volcanic}} = -49.9 \pm 11.8°$; Fig. 7E), we suggest that there was no significant inclination shallowing triggered by depositional compaction in the clastic rocks of the Qijiaojing Formation.

Toudaogou Area

The ChRM direction failed to pass a fold test ($k_g/k_s = 83$) for the Julideneng Formation (McElhinny, 1964), implying the magnetization is "secondary." Moreover, the stepwise unfolding process for this formation illustrated that the precision parameter (k) achieved a maximum value at $3.1\% \pm 4.4\%$ unfolding, which indicates the ChRM is a postfolding magnetization (Figs. 7G–7I; Watson and Enkin, 1993). Since some hydrothermal veins with secondary magnetic minerals can be observed in the samples from the Julideneng Formation, we suggest that magmatic thermal and hydrothermal fluids caused the remagnetization of the Julideneng Formation. There were two main stages of magmatic activiy after the deposition of the Julideneng Formation (ca. 309 Ma). The first stage formed during the late Carboniferous–earliest

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TABLE 1. SITE-MEAN CHARACTERISTIC REMANENT MAGNETIZATION (ChRM) DIRECTIONS FOR THE QIJIAOJING FORMATION FROM THE TURPAN-HAMI BLOCK

Site	S_{lat} (°N)	S_{long} (°E)	Lithology	−/+/G	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	P_{long} (°)	P_{lat} (°)
QJJ1	43°35′31.0″	91°53′2.0″	Siltstone	6/0/0	305.3	–62.2	226.3	–44.6	121.7	6.1	191.3	48.9
QJJ2	43°35′30.0″	91°53′1.0″	Siltstone	7/0/1	304.1	–53.7	214.5	–51.1	88.8	6.0	193.8	60.5
QJJ3	43°35′28.3″	91°52′56.9″	Siltstone	11/0/0	307.2	–38.5	231.2	–53.9	18.4	10.9	177.1	49.8
QJJ4	43°34′37.7″	91°52′36.5″	Sandstone	8/0/0	295.6	–36.8	239.6	–48.1	18.6	13.2	178.2	41.0
QJJ5	43°34′36.8″	91°52′35.7″	Sandstone	8/0/1	294.1	–37.6	234.4	–58.1	41.4	8.1	169.3	49.5
QJJ6	43°32′44.6″	91°52′14.4″	Basalt	8/0/0	290.3	1.7	261.1	–36.6	29.9	10.3	172.5	20.2
QJJ7	43°32′17.1″	91°52′34.9″	Basalt	7/0/1	214	–39.0	265.0	–56.5	70.4	6.7	155.8	27.8
QJJ8	43°32′18″	91°52′34.6″	Basalt	9/1/0	209.7	–42.2	268.2	–60.7	47.6	7.1	149.9	28.4
QJJ12	43°31′58.1″	91°52′19.5″	Basalt	3/0/1	206.8	–23.0	215.0	–50.0	94.4	10.1	194.9	59.6
QJJ13	43°31′57.7″	91°52′19.1″	Basalt	4/1/0	211.4	–19.0	219.7	–45.2	33.9	13.3	196.3	53.9
KE1	43°35′31.4″	91°53′2.0″	Siltstone	7/0/0	294.5	–53.3	217.8	–46.9	35.1	10.3	196.3	56.1
KE2	43°35′28.2″	91°52′57.5″	Siltstone	6/0/0	311.0	–42.6	235.3	–55.1	73.3	7.9	173.1	47.5
KE3	43°35′8.7″	91°52′35.2″	Siltstone	0/0/0	No high-temperature component isolated							
KE4	43°32′43.3″	91°52′14.7″	Basalt	5/0/0	291.5	–20.6	233.3	–37.5	61.8	9.8	191.3	40.6
KE5	43°32′43.1″	91°52′14.9″	Basalt	0/0/0	No high-temperature component isolated							
QJJ9-10; K6	43°32′26.1″	91°52′14.8″	Sandstone	4/0/2	239	–28.3	—	—	21.2	15.5	—	—
					—	—	258.8	–41.1	24.6	14.3	171.3	23.9
KE7	43°31′58.0″	91°52′19.5″	Basalt	6/0/0	205.9	–21.3	217.7	–46.3	65.1	8.4	197.0	55.8
KE8-9	43°31′31.1″	91°52′4.5″	Basalt	3/0/1	256.1	–24.5	—	—	7.8	37.1	—	—
					—	—	256.8	–50.7	97.2	9.9	165.5	30.0
Mean				16 (111)	262.5	–41.6	—	—	4.9	18.6	—	—
							236.6	–50.3	33.5	6.5	—	—
Mean of VGPs						$K = 22.1$		$A_{95} = 8.0$			177.3	44.4

Note: S_{lat} and S_{long}—latitude and longitude of the sampling sites; −/+/G—number of ChRM directions that display reversed/normal polarity/great circle when calculating Fisher site means; D_g, I_g and D_s, I_s—declination and inclination in geographic and stratigraphic coordinates, respectively; k, K—the Fisher precision parameter for the mean of directions and virtual geomagnetic poles (VGPs), respectively; α_{95}, A_{95}—the radius of the cone at 95% confidence level for the means of directions and VGPs, respectively; P_{long} and P_{lat}—longitude and latitude of paleomagnetic poles. Positive fold test: (A) McElhinny's test (1964): $N = 16$, $k_s/k_g = 6.81 > F = 1.84$ and 2.38 at 95% and 99% confidence levels, respectively, indicating positive fold test at the 95% and 99% confidence levels; (B) McFadden's test (1990): $N = 16$, $\xi(1)_{in\ situ} = 13.99 > \xi c = 4.66$ and $6.52 > \xi(1)_{tilt\text{-}corrected} = 3.58$ at 95% and 99% confidence levels respectively; $\xi(2)_{in\ situ} = 14.86 > \xi c = 4.66$ and $6.52 > \xi(2)_{tilt\text{-}corrected} = 0.114$ at 95% and 99% confidence levels respectively, indicating positive fold test at the 95% and 99% confidence levels; (C) Watson and Enkin's test (1993): the stepwise unfolding illustrates that the precision parameter (k) achieves the maximum value at 93.1% unfolding (with 95% uncertainties varying from 88.9% to 96.8%); (D) Enkin's test (2003): $DC_{slope} = 0.936 \pm 0.142$ also shows a positive fold test; DC—direction-correction.

Permian (308–295 Ma; Zhu et al., 2018b; Ni et al., 2021; Song et al., 2018), represented by the postcollisional granites commonly intruding the Julideneng Formation in the eastern part of the study area (Figs. 2B–2C). The second stage formed as locally developed basic lavas during the early Permian, without having caused regional remagnetization in the study area (Zhu et al., 2019). Moreover, the hydrothermal veins (filled with K-feldspar and quartz) are genetically related to intermediate-acidic magmatic rocks. Thus, we believe that the late Carboniferous postcollisional granitoids contributed to the remagnetization in the eastern part of the study area. The nearest granite intrusion to the sampling sites (~2 km away; Fig. 2B) was emplaced ca. 300 Ma, implying that the remagnetization likely took place during this time.

The fold pattern of the Julideneng Formation had been shaped before the deposition of the Kulai Formation and then tilted 27° toward the SSW (165°) according to the lower stratigraphic occurrence of the Kulai Formation owing to later deformation (Fig. S1H). Hence, the primary appearance of the fold needs to be restored, and the ChRM directions of the Julideneng Formation should be corrected. The corrected paleomagnetic data are listed in Table 2 and shown in Figures 7J and 7K (the detail correction method is available in Supplemental Text S3). The corrected ChRM directions still indicate a negative fold test ($k_g/k_s = 82.6$) (McElhinny, 1964). The Watson and Enkin (1993) test gave an optimal concentration at 3.9% ± 3.9% unfolding percent (Fig. 7L) with a corrected optimal concentration direction of $D_g = 235.3°$, $I_g = -55.5°$, $k = 181.7$, $\alpha_{95} = 3.8°$ (Figs. 7J and 7K). In addition, the corrected ChRM directions of the Julideneng Formation are all reverse polarity and consistent with the Kiaman reversed superchron.

Liudaogou Area

The ChRM direction for the Dananhu Formation in the Liudaogou area failed to pass a fold test (McFadden, 1990): $\xi(2)_{in\ situ} = 0.3376$, $\xi(2)_{tilt\text{-}corrected} = 2.850$; critical value ξC at 95% = 2.609, indicating the magnetization should be interpreted as "secondary." In addition, the stepwise unfolding of the Dananhu Formation showed that the k value is maximum at ~0% unfolding (Fig. S4; Watson and Enkin, 1993), indicating a postfolding magnetization. The optimal concentration direction is $D_g = 252.6°$, $I_g = -46.6°$, with $k_s = 15.8$, $\alpha_{95} = 19.8°$, which is not distinct from the characteristic remanence directions of the Qijiaojing and Julideneng Formation at 95% confidence level (Fig. 7J). Considering wide distribution of the late Carboniferous postcollisional granites in the Liudaogou area (Fig. 2C), it can be concluded that the remagnetization of the Dananhu Formation should also have taken place during the late Carboniferous, related to the magmatic thermal and fluid activity.

Consequently, it can be inferred that the ChRM directions of specimens from the Qijiaojing, Julideneng, and Dananhu Formations most likely represent late Carboniferous magnetization. The high consistency of these ChRM directions might indicate no relative tectonic rotations between these sampling localities since the late Carboniferous. Only a few stable ChRMs were obtained from the Dananhu Formation, and they produced a mean direction with significant error ($\alpha_{95} = 19.8°$), which will not be discussed below. The N-dependent A_{95} envelope tests of Deenen et al. (2011) yielded positive results for the Qijiaojing ($A_{95min} = 1.87° < A_{95} = 3.97° < A_{95max} = 4.4°$, $N = 104$; Figs. S3A–S3B; Table S3) and Julideneng Formations ($A_{95min} = 2.43° < A_{95} = 3.10° < A_{95max} = 6.64°$, $N = 54$; Figs. S3C–S3D; Table S3), which indicate a distribution of ChRM directions typically adequate to average out the paleosecular variation (PSV). Thus, two high-quality late Carboniferous paleomagnetic poles were first obtained for the Turpan-Hami block: 44.4°N, 177.3°E, $K = 22.1$, $A_{95} = 8.0°$ (for ca. 313–308 Ma) and 47.8°N, 173.9°E, $K = 116.0$, $A_{95} = 4.8°$ (for ca. 300 Ma), which were calcu-

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TABLE 2. SITE-MEAN CHARACTERISTIC REMANENT MAGNETIZATION (ChRM) DIRECTIONS FOR THE JULIDENENG FORMATION FROM THE TURPAN-HAMI BLOCK

Site	S_{lat} (°N)	S_{long} (°E)	Lithology	−/+	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	P_{long} (°)	P_{lat} (°)
ChRM directions of the Julideneng Formation before correction*												
TGD1	43°9′29.0″	93°49′22.0″	Sandstone	6/0	196.7	−40.8	—	—	48.0	9.8	233.6	65.9
					—	—	197.4	40.3	38.2	11.0	—	—
TGD2	43°9′28.6″	93°49′21.5″	Sandstone	6/0	200.4	−44.2	—	—	34.8	11.5	223.2	66.1
					—	—	199.9	39.0	33.3	11.8	—	—
TGD3	43°9′28.3″	93°49′20.9″	Sandstone	6/0	211.5	−42.2	—	—	45.3	10.1	209.6	58.1
					—	—	40.6	−77.2	39.9	10.7	—	—
TGD4	43°9′28.1″	93°49′20.1″	Sandstone	6/0	212.1	−36.6	76.5	−86.4	32.6	11.9	214.0	54.8
TGD5	43°9′27.9″	93°49′19.9″	Sandstone	6/0	212.3	−37.5	—	—	48.0	9.8	213.0	55.2
					—	—	14.8	−78.7	56.8	9.0	—	—
TGD6	43°9′27.7″	93°49′19.5″	Sandstone	4/0	212.7	−41.1	68.9	−69.6	166.9	7.1	209.3	56.7
TGD7-8	43°9′26.9″	93°49′19.9″	Sandstone	9/0	221.7	−39.7	—	—	45.7	7.7	201.2	49.9
					—	—	336.8	−68.0	68.9	6.2	—	—
TGD9	43°9′34.4″	93°49′29.6″	Sandstone	6/0	209.4	−40.2	—	—	81.2	7.5	214.2	58.4
					—	—	120.0	−70.1	62.8	8.5	—	—
TGD10-11	43°9′35.8″	93°49′30.9″	Sandstone	5/0	207.3	−39.2	—	—	56.4	10.3	217.9	59.1
					—	—	66.5	−80.5	39.5	12.3	—	—
Mean				9 (54)	209.4	−40.4	—	—	181.7	3.8	—	—
							151.8	−81.5	2.3	44.4	—	—
Mean of VGPs						$K = 136.9$		$A_{95} = 4.4$			214	58.5
Corrected ChRM directions of the Julideneng Formation according to the lower bedding attitude (165°∠27°) of the Kulai Formation†												
TGD1	43°9′29.0″	93°49′22.0″	Sandstone	6/0	220.1	−61.0	—	—	48.0	9.8	172.0	60.7
					—	—	215.7	42.4	38.2	11.0	—	—
TGD2	43°9′28.6″	93°49′21.5″	Sandstone	6/0	228.8	−62.4	—	—	34.8	11.5	165.6	55.1
					—	—	218.8	39.8	33.3	11.8	—	—
TGD3	43°9′28.3″	93°49′20.9″	Sandstone	6/0	239.5	−56.1	—	—	45.3	10.1	171.1	44.9
					—	—	33.9	−79.6	39.9	10.7	—	—
TGD4	43°9′28.1″	93°49′20.1″	Sandstone	6/0	234.8	−51.2	65.9	−86.4	32.6	11.9	179.7	45.9
TGD5	43°9′27.9″	93°49′19.9″	Sandstone	6/0	235.8	−51.9	—	—	48.0	9.8	178.3	45.5
					—	—	342.1	−79.6	56.8	9.0	—	—
TGD6	43°9′27.7″	93°49′19.5″	Sandstone	4/0	239.6	−54.7	54.6	−69.6	166.9	7.1	172.8	44.1
TGD7-8	43°9′26.9″	93°49′19.9″	Sandstone	9/0	247.3	−49.5	—	—	45.7	7.7	173.7	36.0
					—	—	335.9	−71.1	68.9	6.2	—	—
TGD9	43°9′34.4″	93°49′29.6″	Sandstone	6/0	235.1	−55.4	—	—	81.2	7.5	174.3	47.6
					—	—	110.8	−64.5	62.8	8.5	—	—
TGD10-11	43°9′35.8″	93°49′30.9″	Sandstone	5/0	231.8	−55.4	—	—	56.4	10.3	176.1	50.1
					—	—	76.4	−76.8	39.5	12.3	—	—
Mean				9 (54)	235.3	−55.5	—	—	181.7	3.8	—	—
							184.2	−85.2	2.2	45.5	—	—
Mean of VGPs						$K = 116.0$		$A_{95} = 4.8$			173.9	47.8

Note: See Table 1 for abbreviation and symbol definitions.

*Fold tests for the sites from the Julideneng Formation are negative according to McElhinny (1964) ($k_g/k_s = 83.0$) with a test statistic maximized at ~3.1% unfolding (with uncertainties ranging from −1.0% to 7.5%) (Watson and Enkin, 1993).

†The fold test is negative according to McElhinny (1964) ($k_g/k_s = 82.6$) for the sites from Julideneng Formation with a test statistic maximized at ~3.9% unfolding (with uncertainties ranging from 0.1% to 7.8%) (Watson and Enkin, 1993).

lated on site-mean VGPs and are reliable for tectonic reconstructions.

Closure of the Western Paleo-Asian Ocean

The North or South Tianshan suture, as the final closure position of the western Paleo-Asian Ocean (Charvet et al., 2011; Xiao et al., 2018; Han and Zhao, 2018), runs east to west in general, implying that the assembly of the Junggar-Turpan-Hami and Tarim blocks might have been a result of an approximate north-south continent-continent collision. After evaluating the previous paleomagnetic data according to the criteria of Meert et al. (2020), the available late Carboniferous to early Permian paleomagnetic data (R value ≥ 4) of blocks on both sides of Tianshan sutures (Siberia, Mongolia-Tuva, South Mongolia, Baltica, Junggar, West Junggar, Turpan-Hami, Yili, and Tarim blocks and Aqishan-Yamansu belt) are listed in Table 3 with the paleopoles shown in Figure 8. In addition, the corresponding paleolatitudes of each block were calculated as the reference point at the mean sampling location (93.0°E, 43.4°N). So far, seven early Permian (Cheng et al., 1983; Bai et al., 1987; Sharps et al., 1989; Fang et al., 1991; Xu et al., 2013b; Wei et al., 2020) and five late Carboniferous (Bai et al., 1987; Meng, 1991; Gilder et al., 1996; Fang et al., 2006; Sun et al., 2014) paleomagnetic poles for the Tarim block have been presented. Most of these are not distinguished from each other at 95% confidence. The mean early Permian and late Carboniferous paleomagnetic poles of these results are at 56.7°N, 181.5°E with $A_{95} = 5.3$ and 59.3°N, 176.9°E with $A_{95} = 6.3$, respectively, showing no significant difference between the late Carboniferous and early Permian poles.

The late Carboniferous paleopoles of the Tarim and Baltica blocks are very close to the ca. 310 Ma and ca. 300 Ma poles of the Turpan-Hami block, which might imply that the late Carboniferous relative paleogeographic position between the three blocks was similar to the current version. Although the late Carboniferous poles for the Turpan-Hami, Tarim, and Baltica blocks are distinct from the coeval poles of the other blocks, these paleomagnetic poles are roughly distributed along small circles centered at the average sampling location (Fig. 8). The paleolatitudes of the Siberia, Mongolia-Tuva, South Mongolia, Baltica, Tarim, Turpan-Hami, West Junggar, and Yili blocks, the Aqishan-Yamansu belt, and Kazakhstan were highly consistent during the late Carboniferous. The latitudinal movements of the Tarim block (on the south side of the Paleo-Asian Ocean) are estimated as 3.9° ± 7.1° and ranging from 3.2° ± 5.8° to 6.9° ± 7.5° compared to the Junggar block and Turpan-Hami block (on the north side of the Paleo-Asian Ocean), respectively, since the late Carboniferous, which are indistinguishable within paleomagnetic errors. These results indicate that the Turpan-Hami block and Junggar block might have been welded with the Tarim block due to the closure of ancient Tianshan oceans, forming a large continent consisting of the Siberia, Mongolia-Tuva, South Mongolia,

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TABLE 3. SUMMARY OF LATE CARBONIFEROUS–EARLY PERMIAN PALEOMAGNETIC POLES OBTAINED FROM THE TURPAN-HAMI BLOCK AND ADJACENT BLOCKS

Blocks	Age (Ma)	S_{lat} (°N)	S_{long} (°E)	$N(n)$	P_{lat} (°N)	P_{long} (°E)	A_{95} (dm/dp)	$D \pm \Delta D$ (°)	Paleolatitude (°N)	R value	Reference
Siberia	C2 (ca. 320)	—	—	—	29.0	158	8.0		37.0 ± 6.5	—	Metelkin et al. (2012)
(APWP)	C2 (ca. 300)	—	—	—	35.0	160.0	8.0		38.8 ± 6.3	—	
Baltica	C2 (ca. 310)	—	—	—	39.7	163.1	4.8		39.0 ± 3.8	—	Torsvik et al. (2012)
(APWP)	C2 (ca. 300)	—	—	—	44.2	166.6	1.6		38.8 ± 1.3	—	
Mongolia-Tuva	C2 (ca. 315)	106.6	47.8	20(139)	43.5	355.9	2.9		24.0 ± 3.0	123F5□7(6)	Ren et al. (2021)
South Mongolia	C2 (ca. 316)	101.0	43.1	(42)	32.2	154.6	7.8		41.2 ± 5.9	123F□□7(5)	Kovalenko and Chernov (2008)
West Junggar	C2 (ca. 320–304)	46.5	83.0	12(107)	70.7	209.0	5.0		32.9 ± 4.4	123F5□7(6)	Yi et al. (2015)
	C2	46.7	86.1	15	70.0	299.0	6.4		25.0 ± 6.6	12□F5□7(5)	Li et al. (1991)
Junggar	C2	43.8	87.8	15	72.0	7.0	7.4	335 ± 5	42.0 ± 5.5	12□F5□7(5)	Li et al. (1991)
	P1	43.8	87.6	4(30)	79.5	36.6	6.8	347.4 ± 11	48.5 ± 4.6	123□5□7(5)	Choulet et al. (2011)
Kazakhstan	C2–P1 (305–275)	48.1	75.6	18(102)	42.2	178.8	4.0		30.1 ± 3.7	□23F567(6)	Abrajevitch et al. (2008)
Yili	C2 (ca. 310)	43.3	83.3	15(98)	68.6	290.6	6.1	352.0 ± 8	22.8 ± 6.5	123F□□7(5)	Wang et al. (2007)
	P1 (ca. 270)	—	—	11	81.5	256.5	10.9	1.1 ± 10	35.2 ± 9.2	123F5□7(6)	Zhu et al. (2018a)
Turpan-Hami block (THB)	C2 (ca. 310)	43.6	91.9	16(111)	44.4	177.3	8.0	56.7 ± 7.2	32.1 ± 7.1	123F5□7(6)	This study
	C2 (ca. 300)	43.2	93.8	9(54)	47.8	173.9	4.8	55.1 ± 4	35.9 ± 4.0	123□5□7(5)	
	P1	—	—	10(98)	75.7	276.3	5.7	358.4 ± 6	29.1 ± 5.4	123F5□7(6)	Zhu et al. (2019)
Aqishan-Yamansu belt (AYB)	321–318	41.9	93.7	13(89)	77.2	301.5	5.5		31.9 ± 4.9	123□5□7(5)	Xu et al. (2021)
	ca. 313	41.9	93.7	5(27)	84.6	38.3	11.6	353.5 ± 10	46.3 ± 8.1	1□3F5□7(5)	
Tarim	C2	41.0	79.0	4(44)	52.2	179.5	8.6		34.8 ± 7.3	123□5□7(5)	Bai et al. (1987)
	C2	—	—	(15)	59.0	185.7	6.6		34.8 ± 5.6	1□3□5□7(4)	Meng (1991)
	C2	37.0	79.0	15	64.6	166.5	6.3		45.2 ± 4.5	□2□F5□7(4)	Gilder et al. (1996)
	C2	39.9	80.2	8	62.0	184.5	12.6		36.7 ± 10.3	□2□F5□7(4)	Fang et al. (2006)
	C2	—	—	25(250)	57.2	167.0	6.1		43.3 ± 4.5	123F5□7(6)	Sun et al. (2014)
	C2 (Mean)	—	—	5*	59.3	176.9	6.3	36.2 ± 6	39.1 ± 4.9	—	—
	P1	40.8	79.8	24(290)	62.8	191.5	3.2		34.2 ± 2.7	12□□5□7(4)	Cheng et al. (1983)
	P1	40.8	79.8	26(317)	56.5	190.1	4.3		31.6 ± 3.9	12□□5□7(4)	Bai et al. (1987)
	P1	40.7	79.7	21(145)	54.5	172.3	4.0		39.6 ± 3.1	12□F5□7(5)	Sharps et al. (1989)
	P1	40.6	79.5	5(27)	61.7	185.2	6.8		36.1 ± 2.7	1□□F5□7(4)	Fang et al. (1991)
	P1	40.6	79.5	11(77)	53.9	178.5	3.2		35.6 ± 2.7	12□F5□7(5)	
	P1	38.7	79.7	9(62)	53.2	187.5	12.3		31.1 ± 11.2	12□□5□7(4)	Xu et al. (2013b)
	P1	40.7	79.6	51(413)	50.1	170.5	3.6		38.9 ± 2.8	123F5□7(6)	Wei et al. (2020)
	P1 (Mean)	—	—	7*	56.7	181.5	5.3	37.3 ± 5	35.8 ± 4.4	—	—

Note: S_{lat} and S_{long}—latitude and longitude of the sampling sites; $N(n)$—number of sites (specimens) used for Fisher statistics; P_{lat} and P_{long}—latitude and longitude of paleomagnetic poles; $D \pm \Delta D$—paleomagnetic declination with the error shown in Figures 9B and 9D at the corresponding sampled location; Paleolatitude—calculated as the reference point at 43.4°N, 93.0°E; R value—seven criteria of paleomagnetic data (number of criteria met) after Meert et al. (2020), where F indicates a positive fold test, and □ indicates a failure to fulfill the corresponding criterion; paleomagnetic data with R values <4 were excluded. APWP—apparent polar wander path.
*Number of paleomagnetic poles used for calculating mean.

Baltica, West Junggar, Junggar, Turpan-Hami, Kazakhstan, Yili, and Tarim blocks and the Aqishan-Yamansu belt. As for the eastern section of the Paleo-Asian Ocean, the significant difference in contemporaneous paleolatitudes between the Dunhuang–Alxa–North China block and the Mongolia–Tova–South Mongolia–Xing'an belt indicate that the east Paleo-Asian Ocean was still open during the late Carboniferous–early Permian (Xu et al., 2019, 2021; Zhang et al., 2021; Ren et al., 2022). The tectonic framework might have lasted until the late Permian, which coincides with the late Paleozoic paleobiogeographic patterns (Xu et al., 2019). Thus, two schematic paleogeographic reconstruction maps of the Paleo-Asian Ocean for the late Carboniferous (ca. 300 Ma) and early Permian (ca. 280 Ma) are suggested after Xu et al. (2019, 2021) in Figures 9A and 9C.

The closure time of the western Paleo-Asian Ocean is essentially in agreement with the aforementioned geologic evidence. First, the North Tianshan suture may represent the terminal stage of development of the Paleo-Asian Ocean, based on the youngest ophiolitic rocks (Bayingou ophiolite with an age of ca. 325 Ma) in the Tianshan-Jungar region (Xu et al., 2006; Dong et al., 2010), and a reliable age of the stitching pluton (undeformed postcollisional granodiorite) within this suture, which was reported as late Carboniferous (316 ± 3 Ma; Han et al., 2010); these might suggest that the ancient North Tianshan Ocean was closed during the period 325–316 Ma. Second, there is an extensive and apparent angular unconformity, named the Shanshan movement (Huang and Zhu, 1986), between the Lower Carboniferous and Upper Carboniferous sections in the Tianshan region. The strata across the unconformity have tremendously different sedimentary facies, paleontological features, and structural deformation, which might imply the closure of the Tianshan Oceans (Li et al., 2009b). Third, postcollisional granodiorites in the late Carboniferous to Permian (ca. 320–270 Ma) commonly intruded in the Tianshan orogenic belts (Han et al., 2010, 2011; Chen et al., 2011; Xu et al., 2013c; Du et al., 2018), well constraining the upper limit of the closure time for the western Paleo-Asian Ocean. Finally, the mixture of Tethyan and Boreal faunas occurred in the Yili area during the early Carboniferous, implying the Tianshan Oceans had become a narrow marine biogeographic migration pathway at that time. Subsequently, a late Carboniferous mixed Angaran and Euramerican flora took over both in the Junggar and Tarim blocks (Ouyang et al., 1994; Wang et al., 2017) as a result of the closure of the western Paleo-Asian Ocean (Figs. 9A–9B).

Although the ancient main Tianshan oceans had been closed since the late Carboniferous, the paleolatitudinal difference (9.9° ± 8.0°) between the Junggar block and the Turpan-Hami block can still be observed in the late Carboniferous (Fig. 8). Note that remnant seas still existed within northern Xinjiang during

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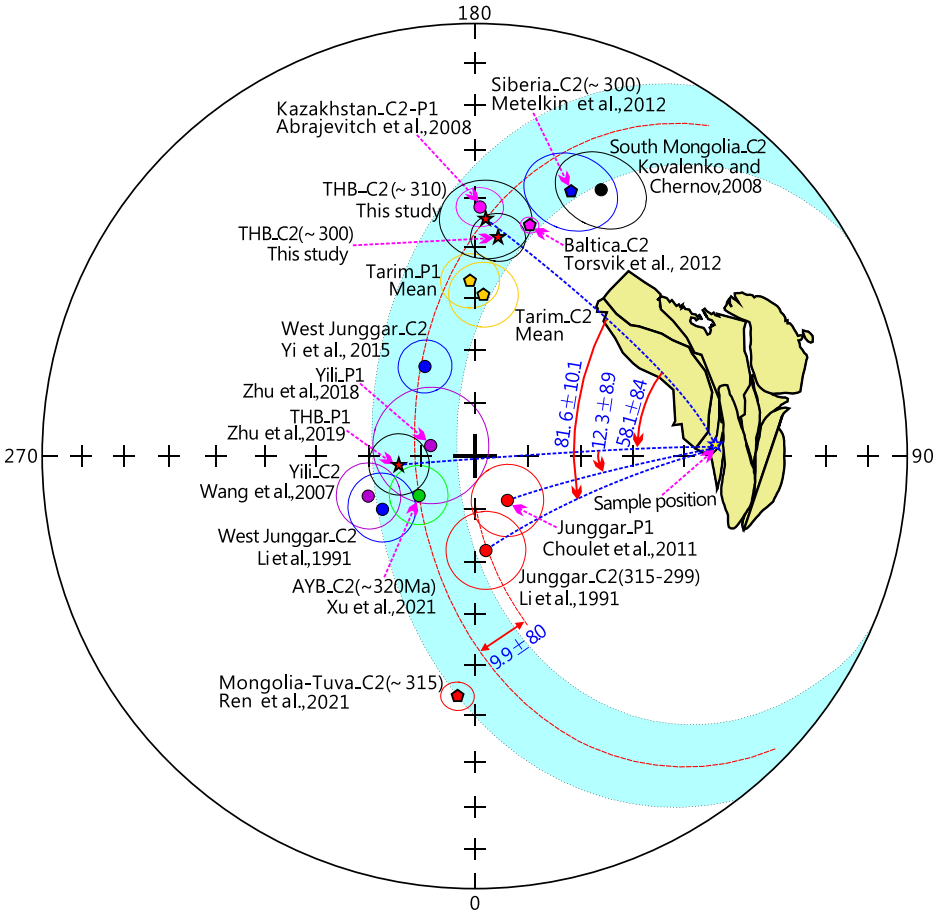


Figure 8. Equal-area projection showing the available late Carboniferous–early Permian paleomagnetic poles for the blocks on both sides of the Tianshan sutures. The corresponding values and references are listed in Table 3. The small circle passing over the late Carboniferous (ca. 310 Ma) pole for the Turpan-Hami block (THB) is fit at the mean sampling location (93.0°E, 43.4°N) with a paleolatitude of 32.1° ± 7.1°. Red stars and orange pentagons represent the paleomagnetic poles of the Turpan-Hami block and the mean poles of the Tarim block, respectively. Blue, purple, pink, red, green, and black circles denote the poles for the West Junggar block, Yili block, Kazakhstan block, Junggar block, Aqishan-Yamansu belt, and South Mongolia, respectively. Blue, pink, and red pentagons denote the poles for the Siberia, Baltica, and Mongolia-Tuva blocks, respectively.

the late Carboniferous to early Permian, which is supported by the simultaneous marine strata distributed in Bogda and southwest Tianshan (Figs. 9B and 9D; Wartes et al., 2002; XGSC, 2007; Wu et al., 2008; Xu et al., 2013a; Wang et al., 2013). The deepening sedimentary environments and bimodal volcanic rocks were widely distributed along the Bogda belt during the late Carboniferous–early Permian (Gu et al., 2001; XGSC, 2007; Shu et al., 2011; Chen et al., 2011; Wali et al., 2018), and the remnant Bogda marine basin was likely exhausted during the middle to late Permian (Zhu et al., 2019). Thus, the latitudinal movement of 9.9° ± 8.0° (1100 ± 890 km) between the Turpan-Hami block and Junggar block should indicate to the closure of the Bogda marine basin and crustal shortening between these two blocks since the late Carboniferous (Huang et al., 2004b). In addition, the discrepancies among the late Carboniferous poles of different blocks within northern Xinjiang indicate that some significant relative movements between these blocks have taken place since the late Carboniferous along some large-scale strike-slip shearing faults (Figs. 9B and 9D; Wang et al., 2007; Charvet et al., 2011; Choulet et al., 2012).

Clockwise Rotation of the Turpan-Hami Block with Scissor-Style Shrinking of the Bogda Basin

An early Permian paleomagnetic pole for the Turpan-Hami block has been reported as 75.7°N, 276.3°E, A_{95} = 5.7° (Zhu et al., 2019). Thus, we can obtain the first apparent polar wander path (including ca. 310 Ma, ca. 300 Ma, and early Permian poles) of the Turpan-Hami block. There was no significant difference between the ca. 310 Ma and ca. 300 Ma poles of the Turpan-Hami block at 95% confidence, indicating that the Turpan-Hami block did not experience prominent latitudinal motion or rotation during the ca. 310–300 Ma period. However, the late Carboniferous poles of the Turpan-Hami block are significantly different from the early Permian pole, which reveals a sizeable clockwise rotation of ~58° for the Turpan-Hami block during the early part of the early Permian (Fig. 8). In addition, the clockwise rotations of the Turpan-Hami block are estimated to be 81.6° ± 10.1° and 12.3° ± 8.9° compared with the Junggar block since the late Carboniferous and early Permian, respectively, indicating the main relative rotation occurred during the early part of the early Permian. Coincidentally, the Bogda basin between the Junggar block and Turpan-Hami block (Figs. 9B and 9D) was characterized by a scissor-like shrinking during the late Carboniferous to early Permian with marine regression from northeast to southwest (Gu et al., 2001; XGSC, 2007; Wu et al., 2008; Xu et al., 2013a; Wang et al., 2013). It is reasonable to assume that the scissor-like convergence should have resulted from the relative rotation between the Junggar block and Turpan-Hami block. Meanwhile, early Permian (298–285 Ma) magmatic rocks and strike-slip pull-apart basins are widely distributed along the periphery of the Turfan-Hami Basin (Gu et al., 2001; Chen et al., 2011; Wali et al., 2018; Mao et al., 2014; Ni et al., 2021), which should relate to crustal thinning owing to the sizeable clockwise rotation of the Turpan-Hami block.

It has been argued that sizeable counterclockwise rotation of the whole region between the South Tianshan belt and Erqishi fault took place between the late Carboniferous and late Permian, resulting in eastward lateral displacements (Wang et al., 2007; Xu et al., 2021). However, this study revealed sizeable clockwise rotation of the Turpan-Hami block during the early part of the early Permian, indicating that the regional eastward displacement occurred after this period. Although the main oceanic basin of the western Paleo-Asian Ocean had been closed since the late Carboniferous, there were still some remnant seas (e.g., the Bogda basin). We argue that some relative rotations between the blocks within the western Altaids might have occurred first with the scissor-like closure of the remnant seas after the main western Paleo-Asian Ocean was closed, and then the large eastward lateral displacements occurred in response to the tectonic stress field caused by the east-west dif-

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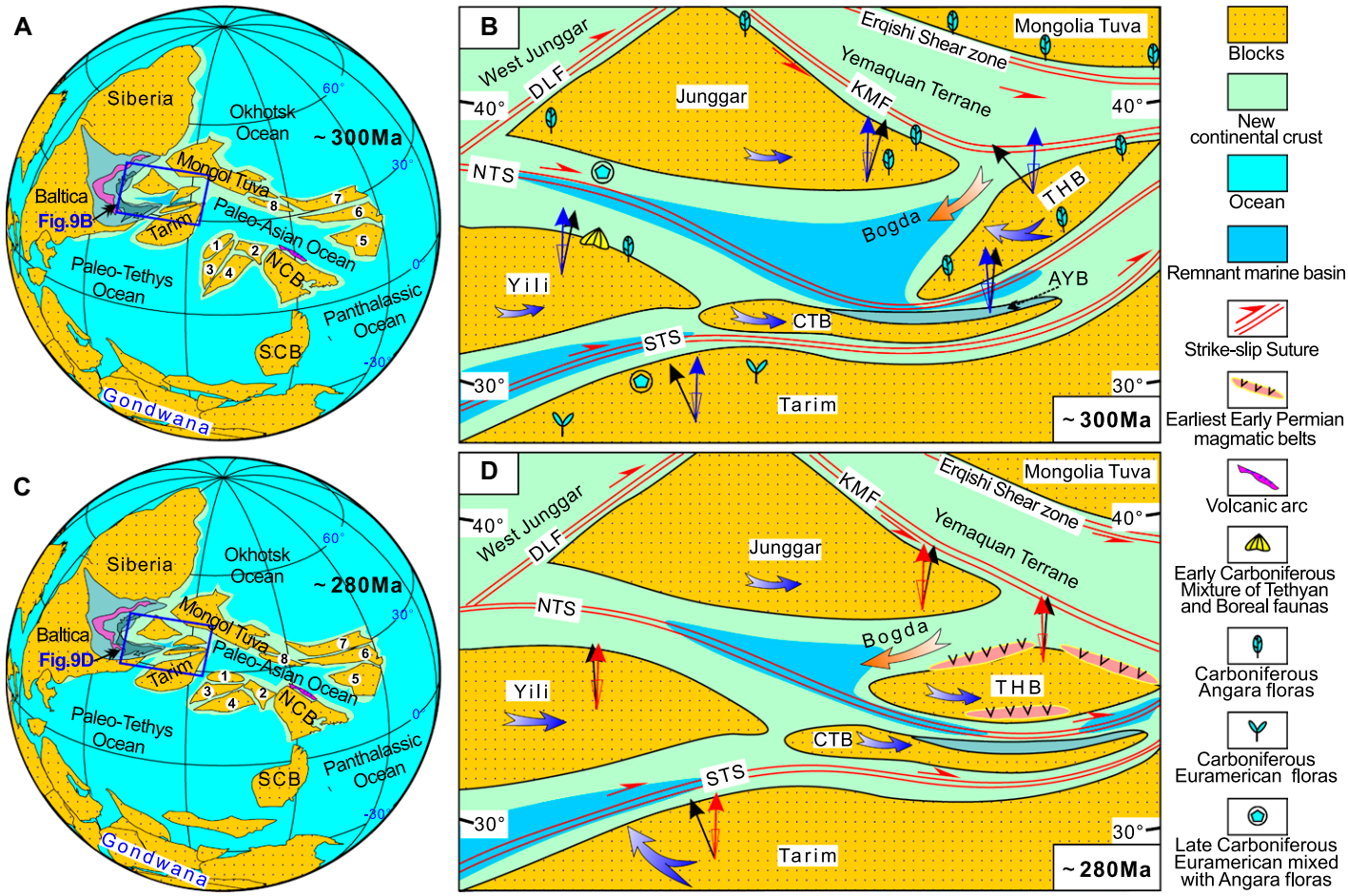


Figure 9. (A, C) Schematic paleogeographic reconstruction maps of Paleo-Asian Ocean for (A) late Carboniferous and (C) early Permian (modified from Xu et al., 2019, 2021). (B, D) Simplified tectonic maps showing geodynamic evolution of northern Xinjiang at (B) ca. 300 Ma and (D) ca. 280 Ma. Blocks and terranes: AYB—Aqishan-Yamansu belt; CTB—Central Tianshan block; NCB—North China block; SCB—South China block; THB—Turpan-Hami block; 1—southern Beishan; 2—Alxa; 3—Dunhuang; 4—Qaidam; 5—Songliao; 6—Xing'an; 7—Erguna; 8—South Mongolia. Abbreviations for sutures and faults are identical to those in Figure 1. Bold blue and red arrows illustrate directions of block motions and marine regression, respectively. Thin blue and red arrows indicate the paleomagnetic declinations with the errors at corresponding sampled locations during late Carboniferous and early Permian, respectively, and black arrows show the present-day north of these areas. Distribution of remnant marine basins in northern Xinjiang is based on Wu et al. (2008), Wang et al. (2013), and Xu et al. (2013a). The post-Permian deformation is considered after Choulet et al. (2013). Distribution points of Permian–Carboniferous floras are based on data from Yao (1991), Wu (1993), Ouyang et al. (1994), Zhang and Wang (1996), Zhu et al. (1998a), Wang et al. (2017), and Xu et al. (2021). The earliest early Permian magmatic belts are based on XGSC (2007), Gu et al. (2001), Chen et al. (2011), Mao et al. (2014), Wali et al. (2018), and Ni et al. (2021).

ference in closure time of the Paleo-Asian Ocean (Xu et al., 2019, 2021). The Paleozoic structures were further reactivated in response to the far-field effects of the diachronous collision of the Tibetan blocks since the Triassic, which was characterized by relative rotation between the blocks in northern Xinjiang without significant latitudinal movements during the Mesozoic (Choulet et al., 2013). On the contrary, Cenozoic deformation manifests as significant continental shortening with slight block rotations under the far-field effects of the India-Asia collision (Chen et al., 1991; Huang et al., 2004a, 2004b), ultimately causing the uplift of the Tianshan Range.

CONCLUSION

This study presented new authentic late Carboniferous geochronologic and paleomagnetic results for the Turpan-Hami block. Based on the results of this study and reliable coeval paleomagnetic poles of the adjacent blocks, the following conclusions are summarized: (1) The central oceanic basin of the western Paleo-Asian Ocean (between the Siberian and Tarim blocks) had been closed since the late Carboniferous (ca. 310 Ma) but with remnant seas, and (2) the Turpan-Hami block underwent a sizeable clockwise rotation estimated to be ~58° during the early part of the early Permian with a scissor-like shrinking of the remnant Bogda Sea (Figs. 9B and 9D).

The paleomagnetic results from the Toudaogou and Liudaogou areas may indicate a widespread late Carboniferous remagnetization related to the magmatic thermal events in the northeastern margin of the Turpan-Hami Basin, which needs to be verified in future studies. More high-quality paleomagnetic poles are still required in northern Xinjiang of China to improve understanding of the kinematic history of the western Altaids during the late Paleozoic.

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